



A Two-Graph Refinement of Paulsen's Lollipop Bounds

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Abstract

Let $a_{\mathcal{L}}(n)$ be the maximum number of regions into which n lollipops divide the plane. Paulsen introduced a second obstruction for this problem, based on pairs of circles meeting at obtuse angle, in addition to the stem-direction obstruction of Cutler–Karlsson–Sloane. We recast Paulsen's argument as a weighted problem for two graphs: a K_4 -free graph D of non-close stem pairs and a K_5 -free graph E of non-intriguing circle pairs. For the total number C of pairwise crossings,

$$C \leq 4 \binom{n}{2} + |D| + |E| + |D \cap E|.$$

Paulsen bounds the final term by $|D|$. We keep the overlap term and analyze near-extremal configurations of D and E . This closes all of Paulsen's remaining gaps up to $n = 17$, and also closes $n = 19$:

$$\begin{aligned} & a_{\mathcal{L}}(0), a_{\mathcal{L}}(1), \dots, a_{\mathcal{L}}(17) \\ &= 1, 2, 10, 25, 45, 71, 104, 142, 186, 237, 294, 356, 425, 500, 580, 667, 761, 859, \end{aligned}$$

and

$$a_{\mathcal{L}}(19) = 1076.$$

The same method gives the one-region gaps

$$964 \leq a_{\mathcal{L}}(18) \leq 965, \quad 1193 \leq a_{\mathcal{L}}(20) \leq 1194.$$

1 Introduction and statement

A *lollipop* is a circle together with a half-line attached at one point of the circle and extending radially outward. Equivalently, it is determined by a center $C \in \mathbb{R}^2$, a radius $r > 0$, and a unit vector u , and is the union of

$$\{P : |P - C| = r\} \quad \text{and} \quad \{C + ru + tu : t \geq 0\}.$$

Let $a_{\mathcal{L}}(n)$ be the largest possible number of connected regions into which n lollipops divide the plane.

We work in generic arrangements: there are no tangencies, no triple crossing points, and no crossing occurs at an anchor. This entails no loss for an extremal problem, since a non-generic maximizer can be perturbed without decreasing the number of regions. If C is the total number of crossings between distinct lollipops, then

$$\#\text{regions} = C + n + 1. \quad (1)$$

Thus the region problem is equivalent to maximizing C .

The values $a_{\mathcal{L}}(0), \dots, a_{\mathcal{L}}(4) = 1, 2, 10, 25, 45$ are due to Cutler, Karlsson, and Sloane [1]. Paulsen's recent note proves the exact values for $n = 5, 6, 7, 10$ and gives the best previously published upper bounds for the remaining small cases [2]. The OEIS entry A389624 has been updated accordingly [3].

Our contribution is purely combinatorial once Paulsen's two geometric inputs are accepted. We do not change the known construction; instead we sharpen the upper bound by analyzing how the two extremal obstruction graphs can overlap.

Theorem 1. *The following values are exact:*

$$\begin{aligned} a_{\mathcal{L}}(0), a_{\mathcal{L}}(1), \dots, a_{\mathcal{L}}(17) \\ = 1, 2, 10, 25, 45, 71, 104, 142, 186, 237, 294, 356, 425, 500, 580, 667, 761, 859, \end{aligned}$$

and

$$a_{\mathcal{L}}(19) = 1076.$$

Moreover,

$$964 \leq a_{\mathcal{L}}(18) \leq 965, \quad 1193 \leq a_{\mathcal{L}}(20) \leq 1194.$$

The matching lower bounds in Theorem 1 come from the standard blow-up of Karlsson's optimal four-lollipop configuration. We recall the construction in Appendix B. The rest of the paper proves the upper bounds.

2 Paulsen's two obstructions

We use Paulsen's terminology [2].

Definition 2. A pair of lollipops is *close* if the angle between its two stem directions is at most 90° . A pair is *intriguing* if its two circles either do not intersect, or intersect at angle at most 90° , where the intersection angle is measured after orienting both circles in the same direction.

The following are the geometric inputs. The first close-pair estimate is the stem-angle lemma of Cutler–Karlsson–Sloane, and the intriguing-pair estimates and forced intriguing-pair theorem are due to Paulsen.

Lemma 3 (Cutler–Karlsson–Sloane and Paulsen). *For a generic pair of lollipops:*

- (i) a close pair has at most 5 crossings;
- (ii) an intriguing pair has at most 5 crossings;
- (iii) a pair that is both close and intriguing has at most 4 crossings.

Furthermore, among any four lollipops there is a close pair, and among any five lollipops there is an intriguing pair.

For reference, Paulsen's linear-algebra proof of the last assertion is recalled in Appendix A, with a minor sign-cleanup to handle zero coefficients.

Given an arrangement of n lollipops, define two graphs D and E on $[n]$:

$$\begin{aligned} ij \in D &\iff L_i, L_j \text{ are not close,} \\ ij \in E &\iff L_i, L_j \text{ are not intriguing.} \end{aligned}$$

Lemma 3 implies

$$D \text{ is } K_4\text{-free,} \quad E \text{ is } K_5\text{-free.}$$

Let $N = \binom{n}{2}$, and let c_{ij} be the number of crossings between L_i and L_j . The four pair estimates in Lemma 3 are summarized by

$$c_{ij} \leq 4 + \mathbf{1}_{ij \in D} + \mathbf{1}_{ij \in E} + \mathbf{1}_{ij \in D \cap E}. \quad (2)$$

Summing over all pairs gives

$$C \leq 4N + \sigma(D, E), \quad \sigma(D, E) := |D| + |E| + |D \cap E|. \quad (3)$$

Let $t_r(n)$ denote the number of edges of the balanced complete r -partite Turan graph $T_r(n)$. Since D is K_4 -free and E is K_5 -free,

$$|D| \leq t_3(n), \quad |E| \leq t_4(n).$$

Paulsen additionally uses $|D \cap E| \leq |D|$ and obtains

$$C \leq U_P(n) := 4\binom{n}{2} + 2t_3(n) + t_4(n). \quad (4)$$

Equivalently,

$$C \leq 7\binom{n}{2} - 2\left\lceil \frac{n(n-3)}{6} \right\rceil - \left\lceil \frac{n(n-4)}{8} \right\rceil.$$

We improve (4) by showing that equality, and in many cases near equality, is incompatible with the simultaneous structures of D and E .

3 One-defect cases

The easiest improvements use only the equality case of Turan's theorem.

Proposition 4. *For $n \in \{8, 9, 11, 13\}$,*

$$C \leq U_P(n) - 1.$$

Consequently,

$$a_{\mathcal{L}}(8) = 186, \quad a_{\mathcal{L}}(9) = 237, \quad a_{\mathcal{L}}(11) = 356, \quad a_{\mathcal{L}}(13) = 500.$$

Proof. Equality in (4) would force

$$|D| = t_3(n), \quad |E| = t_4(n), \quad D \subseteq E.$$

Thus $D = T_3(n)$ and $E = T_4(n)$. Since $D \subseteq E$, every part of the complete quadripartite graph E must be contained in one part of the complete tripartite graph D . Equivalently, the part sizes of $T_4(n)$ must be groupable into the part sizes of $T_3(n)$.

For the four values in question this grouping is impossible:

n	$T_3(n)$ part sizes	$T_4(n)$ part sizes
8	3, 3, 2	2, 2, 2, 2
9	3, 3, 3	3, 2, 2, 2
11	4, 4, 3	3, 3, 3, 2
13	5, 4, 4	4, 3, 3, 3

Therefore $\sigma(D, E) \leq 2t_3(n) + t_4(n) - 1$, so $C \leq U_P(n) - 1$. The construction in Appendix B attains the resulting upper bound. $\square$

4 Internal edges over an extremal tripartition

The remaining improvements need a small extremal lemma. Assume first that $D = T_3(n)$, with parts A, B, C of sizes

$$p \geq q \geq r.$$

Write

$$m = |D \setminus E|, \quad x = |E \setminus D|.$$

Thus m counts missing cross-edges of E relative to the tripartition, while x counts internal E -edges inside A, B , and C .

Lemma 5. *Assume $p \geq q \geq r \geq 4$, $D = T_3(n)$, and E is K_5 -free. With m and x as above:*

(a) *if $m = 0$, then*

$$x \leq \left\lfloor p^2/4 \right\rfloor;$$

(b) if $m = 1$, then

$$x \leq \max \left\{ \left\lfloor p^2/4 \right\rfloor, p + q - 2 \right\};$$

(c) if $m = 2$, then

$$x \leq \max \left\{ \left\lfloor p^2/4 \right\rfloor, p + q - 2, 2p + q - 5 \right\}.$$

Proof. Let H_A, H_B, H_C be the internal graphs induced by E on the three parts. We call a part *active* if its internal graph has at least one edge.

First, each H_A, H_B, H_C is triangle-free. Indeed, if, say, H_A contained a triangle, then since $m \leq 2$ and $|B|, |C| \geq 4$, we could choose vertices $b \in B$ and $c \in C$ incident to no missing cross-edge. The triangle together with b and c would span a K_5 in E , a contradiction.

Second, suppose X and Y are two active parts, with third part Z . For every internal edge $e \in H_X$ and every internal edge $f \in H_Y$, some missing cross-edge must have one endpoint in e and one endpoint in f . Otherwise, choosing a vertex of Z incident to no missing cross-edge would make $e \cup f \cup \{z\}$ a K_5 in E .

If $m = 0$, the second observation implies that at most one part is active. Mantel's theorem inside the largest active part gives $x \leq \lfloor p^2/4 \rfloor$.

Let $m = 1$. If only one part is active, the same Mantel bound applies. If two parts X, Y are active, the unique missing cross-edge must join X to Y , say xy . The second observation implies that every edge of H_X contains x , and every edge of H_Y contains y . Hence the two internal graphs are stars, contributing at most $(|X| - 1) + (|Y| - 1) \leq p + q - 2$ edges.

Let $m = 2$. One active part gives the Mantel bound. Three active parts are impossible, since every pair of active parts would require a missing cross-edge between them. Thus suppose exactly two parts X, Y are active, with $|X| = s \geq t = |Y|$. If only one of the two missing cross-edges joins X to Y , the preceding star argument gives at most $s + t - 2 \leq p + q - 2$ internal edges.

It remains to handle the case where both missing cross-edges join X to Y . Write them as a_1b_1 and a_2b_2 , where $a_i \in X$ and $b_i \in Y$, allowing the two missing edges to share an endpoint. The second observation says that for every $e \in H_X$ and $f \in H_Y$, there is an $i \in \{1, 2\}$ with $a_i \in e$ and $b_i \in f$.

If the two missing edges share an endpoint in X , then every edge of H_X contains that endpoint, while every edge of H_Y meets $\{b_1, b_2\}$. Since H_Y is triangle-free, the latter contributes at most $2t - 4$ edges, and H_X contributes at most $s - 1$. The total is at most $s + 2t - 5 \leq 2s + t - 5$. The case of a shared endpoint in Y is even smaller, by symmetry and $s \geq t$.

Finally assume the two missing edges are disjoint. Every edge of H_X meets $\{a_1, a_2\}$, and every edge of H_Y meets $\{b_1, b_2\}$. Classify an edge by which distinguished endpoints it contains. If H_X contains both an edge using only a_1 and an edge using only a_2 , then every edge of H_Y must contain both b_1 and b_2 , so $|E(H_Y)| \leq 1$ and the triangle-free bound $|E(H_X)| \leq 2s - 4$ gives total at most $2s - 3 \leq 2s + t - 5$. Otherwise all edges of H_X contain a common distinguished endpoint, so $|E(H_X)| \leq s - 1$; the same classification gives $|E(H_Y)| \leq 2t - 4$, and the total is at most $s + 2t - 5 \leq 2s + t - 5$. This proves the $m = 2$ bound. $\square$

We also need one standard covering fact.

Lemma 6. *Let $K_{a,b,c,d}$ be a complete quadripartite graph with $a \leq b \leq c \leq d$. To destroy all transversal K_4 's, one must delete at least ab edges. This is sharp.*

Proof. Deleting all edges between the two smallest parts destroys every transversal K_4 , so ab deletions suffice.

For the lower bound, assign weight $1/(cd)$ to every transversal K_4 . The total weight is

$$abcd \cdot \frac{1}{cd} = ab.$$

An edge between parts of sizes s and t lies in the product of the other two part sizes many transversal K_4 's. This product is at most cd , so each deleted edge covers weight at most 1. At least ab deleted edges are necessary. $\square$

5 Two- and three-defect cases

The next propositions apply the preceding lemmas to $\sigma(D, E)$.

Proposition 7. *For $n \in \{12, 14, 16\}$,*

$$C \leq U_P(n) - 2.$$

Consequently,

$$a_{\mathcal{L}}(12) = 425, \quad a_{\mathcal{L}}(14) = 580, \quad a_{\mathcal{L}}(16) = 761.$$

Proof. If $|D| \leq t_3(n) - 1$, then

$$\sigma(D, E) \leq 2|D| + |E| \leq 2(t_3(n) - 1) + t_4(n),$$

so two crossings are already saved.

It remains to consider $D = T_3(n)$. Put $\delta = t_4(n) - t_3(n)$. Since

$$|E| = t_3(n) - m + x, \quad |D \cap E| = t_3(n) - m,$$

we have

$$\sigma(D, E) = 3t_3(n) + x - 2m. \tag{5}$$

Saving two crossings is equivalent to

$$x - 2m \leq \delta - 2.$$

If $m \geq 2$, this follows from $|E| \leq t_4(n)$, since $x \leq \delta + m$ and therefore $x - 2m \leq \delta - m \leq \delta - 2$.

For $m = 0, 1$, Lemma 5 gives the following bounds:

n	(p, q, r)	δ	$m = 0$	$m = 1$
12	(4, 4, 4)	6	4	6
14	(5, 5, 4)	8	6	8
16	(6, 5, 5)	11	9	9

In each row these imply $x \leq \delta - 2$ for $m = 0$ and $x \leq \delta$ for $m = 1$, as required. The lower construction attains the resulting upper bound. $\square$

Proposition 8. For $n \in \{15, 17, 18, 19, 20\}$,

$$C \leq U_P(n) - 3.$$

Consequently,

$$a_{\mathcal{L}}(15) = 667, \quad a_{\mathcal{L}}(17) = 859, \quad a_{\mathcal{L}}(19) = 1076,$$

and

$$964 \leq a_{\mathcal{L}}(18) \leq 965, \quad 1193 \leq a_{\mathcal{L}}(20) \leq 1194.$$

Proof. If $|D| \leq t_3(n) - 2$, then

$$\sigma(D, E) \leq 2(t_3(n) - 2) + t_4(n),$$

which saves four crossings.

Now suppose $|D| = t_3(n) - 1$. If $|E| \leq t_4(n) - 1$ or $|D \cap E| \leq |D| - 1$, then

$$\sigma(D, E) \leq 2t_3(n) + t_4(n) - 3,$$

which is enough. The only dangerous subcase is

$$|E| = t_4(n), \quad D \subseteq E.$$

Then $E = T_4(n)$, and one must delete exactly

$$t_4(n) - (t_3(n) - 1) = (t_4(n) - t_3(n)) + 1$$

edges from $T_4(n)$ while destroying all transversal K_4 's. Lemma 6 rules this out in all required cases:

n	$T_4(n)$ parts	$(t_4 - t_3) + 1$	cover lower bound
15	4, 4, 4, 3	10	12
17	5, 4, 4, 4	13	16
18	5, 5, 4, 4	14	16
19	5, 5, 5, 4	16	20
20	5, 5, 5, 5	18	25

It remains to take $D = T_3(n)$. With $\delta = t_4(n) - t_3(n)$, saving three crossings is equivalent, by

(5), to

$$x - 2m \leq \delta - 3.$$

If $m \geq 3$, then $|E| \leq t_4(n)$ gives $x - 2m \leq \delta - m \leq \delta - 3$. For $m = 0, 1, 2$, Lemma 5 gives:

n	(p, q, r)	δ	$m = 0$	$m = 1$	$m = 2$
15	(5, 5, 5)	9	6	8	10
17	(6, 6, 5)	12	9	10	13
18	(6, 6, 6)	13	9	10	13
19	(7, 6, 6)	15	12	12	15
20	(7, 7, 6)	17	12	12	16

Every row satisfies $x - 2m \leq \delta - 3$. The lower construction matches the resulting upper bound for $n = 15, 17, 19$, and leaves one crossing open for $n = 18, 20$. $\square$

6 Numerical table

Combining Paulsen's exact cases $n = 5, 6, 7, 10$ with Propositions 4, 7, and 8 gives the following table. The final column is the consequence of this note, not the current OEIS status.

n	construction	refined upper	Paulsen upper	conclusion
0	1	1	1	$a_{\mathcal{L}}(0) = 1$
1	2	2	2	$a_{\mathcal{L}}(1) = 2$
2	10	10	10	$a_{\mathcal{L}}(2) = 10$
3	25	25	25	$a_{\mathcal{L}}(3) = 25$
4	45	45	45	$a_{\mathcal{L}}(4) = 45$
5	71	71	71	$a_{\mathcal{L}}(5) = 71$
6	104	104	104	$a_{\mathcal{L}}(6) = 104$
7	142	142	142	$a_{\mathcal{L}}(7) = 142$
8	186	186	187	$a_{\mathcal{L}}(8) = 186$
9	237	237	238	$a_{\mathcal{L}}(9) = 237$
10	294	294	294	$a_{\mathcal{L}}(10) = 294$
11	356	356	357	$a_{\mathcal{L}}(11) = 356$
12	425	425	427	$a_{\mathcal{L}}(12) = 425$
13	500	500	501	$a_{\mathcal{L}}(13) = 500$
14	580	580	582	$a_{\mathcal{L}}(14) = 580$
15	667	667	670	$a_{\mathcal{L}}(15) = 667$
16	761	761	763	$a_{\mathcal{L}}(16) = 761$
17	859	859	862	$a_{\mathcal{L}}(17) = 859$
18	964	965	968	$964 \leq a_{\mathcal{L}}(18) \leq 965$
19	1076	1076	1079	$a_{\mathcal{L}}(19) = 1076$
20	1193	1194	1197	$1193 \leq a_{\mathcal{L}}(20) \leq 1194$

Table 1: Refined bounds through $n = 20$. Paulsen's upper bound is $U_P(n) + n + 1$.

Remark 9 (what remains at $n = 18$ and $n = 20$). For $n = 18$ and $n = 20$, the lower construction

is still one crossing below the refined upper bound. Proving exactness in either case requires saving one more unit in $\sigma(D, E)$. In particular, for $n = 18$, the case $D = T_3(18)$ with parts $(6, 6, 6)$ is not the apparent obstruction: an exhaustive check over the missing cross-edge types gives $x \leq 15$ when $m = 3$, which would already save four crossings in that subcase. The remaining difficulty lies in near-extremal cases with $|D| = t_3(18) - 1$, or analogous near-extremal cases at $n = 20$.

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A Paulsen's forced intriguing pair

We recall Paulsen's proof that among five circles, some pair is intriguing. Suppose, toward a contradiction, that five circles with centers $x_i \in \mathbb{R}^2$ and radii $r_i > 0$ are pairwise non-intriguing. Thus every pair intersects at an obtuse angle, equivalently

$$r_i^2 + r_j^2 < |x_i - x_j|^2 < (r_i + r_j)^2 \quad (i \neq j). \quad (6)$$

For each circle set

$$v_i = \frac{1}{r_i} (1, r_i^2 - |x_i|^2, x_i) \in \mathbb{R}^4,$$

and write vectors in $\mathbb{R}^4$ as triples (α, β, x) , with $x \in \mathbb{R}^2$. Endow $\mathbb{R}^4$ with the symmetric bilinear form

$$\langle (\alpha, \beta, x), (\alpha', \beta', x') \rangle = \frac{1}{2} \alpha \beta' + \frac{1}{2} \beta \alpha' + x \cdot x'.$$

A direct computation gives

$$\langle v_i, v_i \rangle = 1, \quad -1 < \langle v_i, v_j \rangle < 0 \quad (i \neq j).$$

Since five vectors in $\mathbb{R}^4$ are linearly dependent, choose a nonzero linear relation and move the negative terms to the other side:

$$\sum_{i \in P} a_i v_i = \sum_{j \in N} b_j v_j, \quad a_i, b_j > 0,$$

where P and N are nonempty disjoint sets. Replacing the relation by its negative if necessary, assume $|P| \leq 2$. Let $w = \sum_{i \in P} v_i$. Then

$$\left\langle \sum_{i \in P} a_i v_i, w \right\rangle > 0,$$

because $|P| = 1$ is immediate and, if $P = \{1, 2\}$, the expression is $(a_1 + a_2)(1 + \langle v_1, v_2 \rangle) > 0$. On the other hand,

$$\left\langle \sum_{j \in N} b_j v_j, w \right\rangle < 0,$$

since every cross inner product $\langle v_j, v_i \rangle$, with $j \in N$ and $i \in P$, is negative. This contradicts the linear relation. Therefore five pairwise non-intriguing circles cannot exist.

B The lower construction

The standard lower construction blows up the optimal four-llollipop crossing table

	1	2	3	4
1	*	5	7	7
2		*	7	7
3			*	7
4				*

by replacing lollipop i with m_i nearby perturbed copies. Crossings between different clusters are inherited from the table, while pairs inside one cluster contribute 4 crossings. Hence

$$L(m_1, m_2, m_3, m_4) = 5m_1m_2 + 7m_1m_3 + 7m_1m_4 + 7m_2m_3 + 7m_2m_4 + 7m_3m_4 \\ + 4 \sum_{i=1}^4 \binom{m_i}{2}.$$

The lower crossing bound is

$$L(n) = \max_{m_1+m_2+m_3+m_4=n} L(m_1, m_2, m_3, m_4),$$

and the corresponding lower region bound is $L(n) + n + 1$.

n	one optimizing (m_1, m_2, m_3, m_4)	$L(n)$	$L(n) + n + 1$
4	(1,1,1,1)	40	45
5	(1,1,1,2)	65	71
6	(1,1,2,2)	97	104
7	(1,2,2,2)	134	142
8	(1,2,2,3)	177	186
9	(1,2,3,3)	227	237
10	(2,2,3,3)	283	294
11	(2,2,3,4)	344	356
12	(2,2,4,4)	412	425
13	(2,3,4,4)	486	500
14	(2,3,4,5)	565	580
15	(2,3,5,5)	651	667
16	(3,3,5,5)	744	761
17	(3,3,5,6)	841	859
18	(3,3,6,6)	945	964
19	(3,4,6,6)	1056	1076
20	(4,4,6,6)	1172	1193